WEEK 3 · DAY 1

Alarms That Cry Wolf

Weeks 1–2 asked “what does this series look like?”
From today we ask: **“when is a reading WRONG?”** 🚨



ALARM! ALARM!

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Three moves



Meet your alarm

a fixed limit is already a model



Watch it cry wolf

46 eclipse nights, all “alarms”



Give it context

same limit, 4 alarms, all real

⏱️ 10 min hook

🎤 30 min slides

👩🏾‍💻 60 min hands-on

🏁 20 min wrap-up

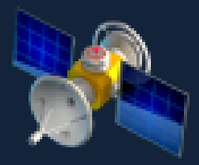
WHY WE'RE HERE

November 2008. NigComSat-1 goes silent.



Solar-array anomaly

Not enough power. The battery couldn't carry it.



Satellite lost

About four years with no Nigerian satellite.



Watch the array

Today's main character: **solar_array_current_a**

PART 1

The alarm you already have 🚨

SURPRISE

Your NOC already does anomaly detection 🎉

```
if reading < limit:
    raise_alarm()          # this is a model!
```

📏

It has a rule

“below 20 A is bad”

🧮

It has a number

someone chose 20

⚖️

It can be wrong

too many alarms, or too few

WHAT ALARM SOFTWARE ALREADY DOES

Four tricks your NOC already uses



Yellow & red limits

Warning first, then alarm. Like the fuel light, then the car stopping.



Persistence

Only alarm if it stays bad for N readings. “Let me see if NEPA really took light.”



Hysteresis

Alarm ON at one level, OFF at another, so it doesn't flap on/off all night.



State-based limits

Different “normal” for different modes: eclipse, recharge, sunlit.

Each trick buys **quiet**… and each one can make you **blind**. Today we test them on data.

THE MOST IMPORTANT QUESTION THIS WEEK

“Where did that number come from?” 🕵🏾

Possible answers 🙈

“The vendor manual said so.”
“We copied it from the old satellite.”
“Oga set it years ago.”
“Nobody knows.”

Good answers ✅

“From the design margin, with this reason.”
“From 6 months of normal data: mean ± 3σ.”
“Reviewed after the last eclipse season.”

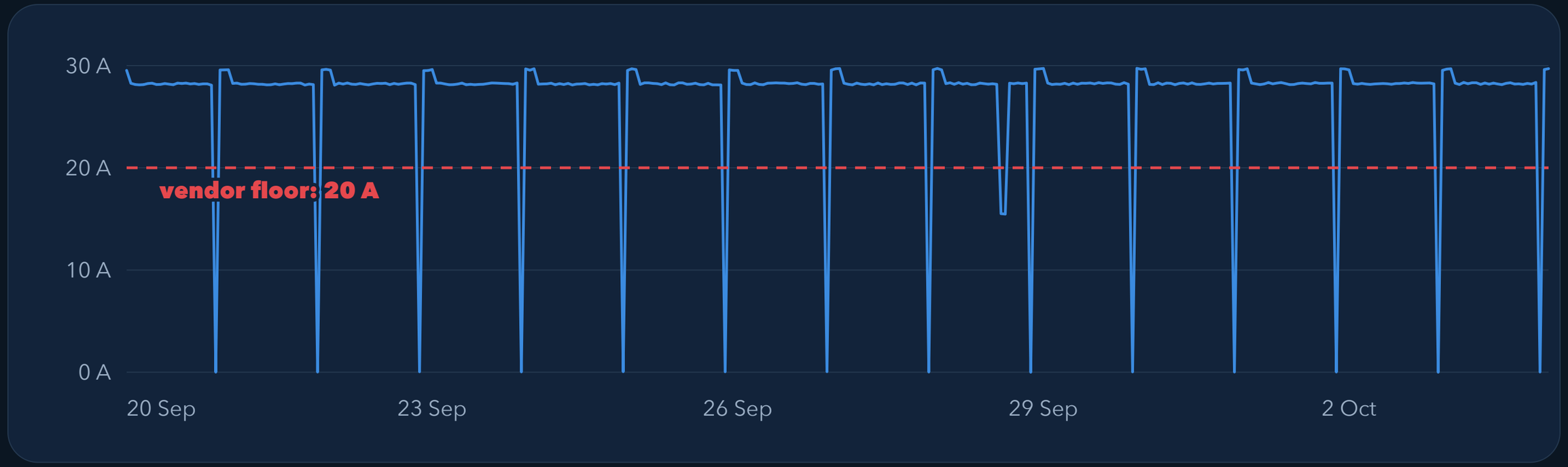
Ask it of **every** limit in your unit. Write the answer down.

PART 2

Watch it cry wolf 🐺

ECLIPSE SEASON

21:00 UTC every night: NEPA takes light 🌘



46 eclipse hours · **1 Sep → 16 Oct** · 🛢️ **runs on battery, like your inverter**

solar-array current, 20 Sep – 3 Oct

THE VENDOR FLOOR: ARRAY CURRENT < 20 A

50 alarms. Only 4 are real.



46 eclipse nights: technically correct, operationally useless

4 real solar-array dropouts hiding among them

ALARM FATIGUE

The car alarm on your street 🚗

It goes off every time a keke passes. Week 1: everyone runs outside. Week 3: nobody even looks.

Then one night, a real thief comes. 🥷 **Nobody moves.**

A NOC that gets 46 false alarms a season learns to **ignore the solar-array alarm**. That's how a NigComSat-1 story starts.



Too many wolves = no shepherd

THE FIX: DON'T WIDEN THE LIMIT. ADD CONTEXT.

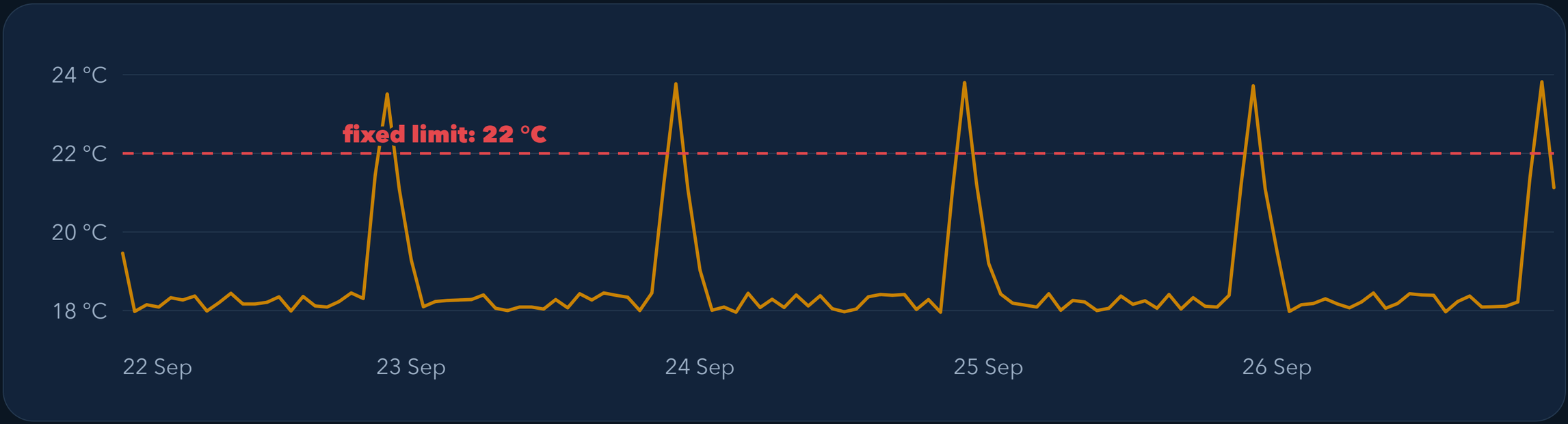
Same limit. Ask one more question first.




```
smart_alarm = (bus["solar_array_current_a"] < 20) & (bus["eclipse_flag"] == 0)
# 50 alarms  →  4 alarms, every one a real dropout
```

SAME TRAP, ANOTHER SENSOR

Battery > 22 °C: 50 alarms, 46 after eclipse



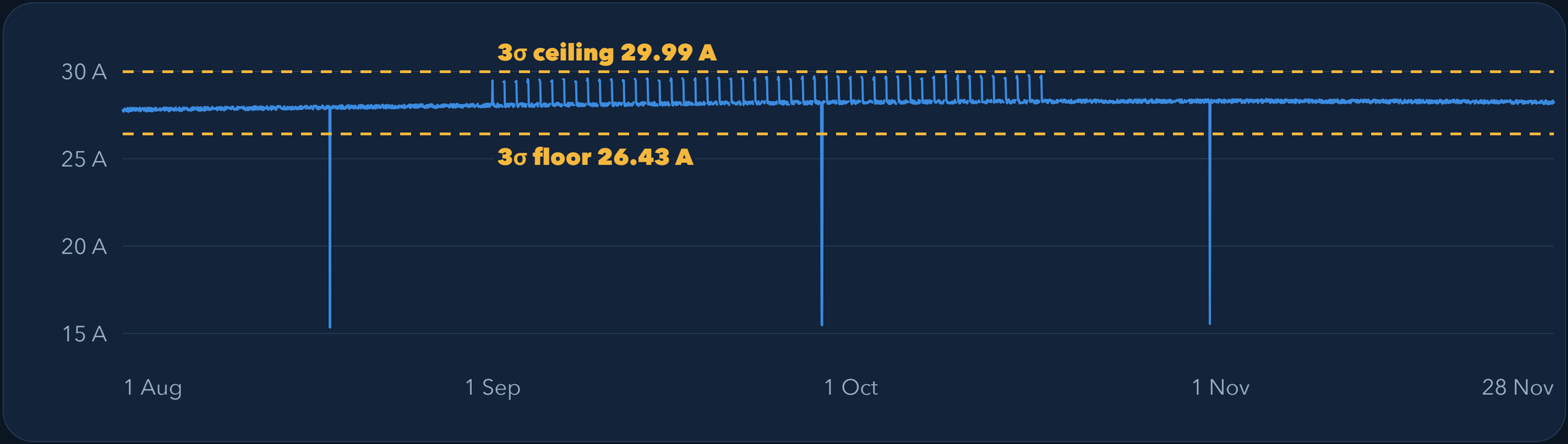
Recharging warms the pack for an hour after every eclipse. **Normal.** Context fix: ignore the hour after eclipse → **4 alarms left, all real.**

PART 3

Where should the limit be? 📏

LET THE DATA CHOOSE THE NUMBER

Normal ± 3σ, from sunlit hours only



mean 28.21 A **σ = 0.594 A** **outside the band: 4 hours = the 4 dropouts**

BUT WHAT IS NORMAL?

The faults you hunt widen your own limits

📦 All sunlit hours

σ = 0.59

Includes the dropouts and the post-eclipse recharge boost. Band 26.4–30.0 A: **wide, quiet**.

🫧 "Clean" hours only

σ = 0.17

Remove every odd hour. Band 27.65–28.66 A: **143 alarms**, 138 of them normal recharge hours. 🫏

Normal needs **context** (sunlit? recharging? eclipse?), not just a mean and a σ. Day 2 builds this properly.

THE RF LINK: RAINING ≠ LINK DOWN

A 3σ rule on C/N can't tell rain from outage

3σ from ALL hours

18

alarms (floor 10.76 dB) · **12 real** · **6 just rain**

3σ from DRY hours only

49

alarms (floor 14.06 dB) · **12 real** · **37 just rain** (76% false)

August had **56** rainy hours and **12** real outages. **Link margin below 0 dB = down.** Rain alone = not a fault.

THE PERSISTENCE TRICK, TESTED

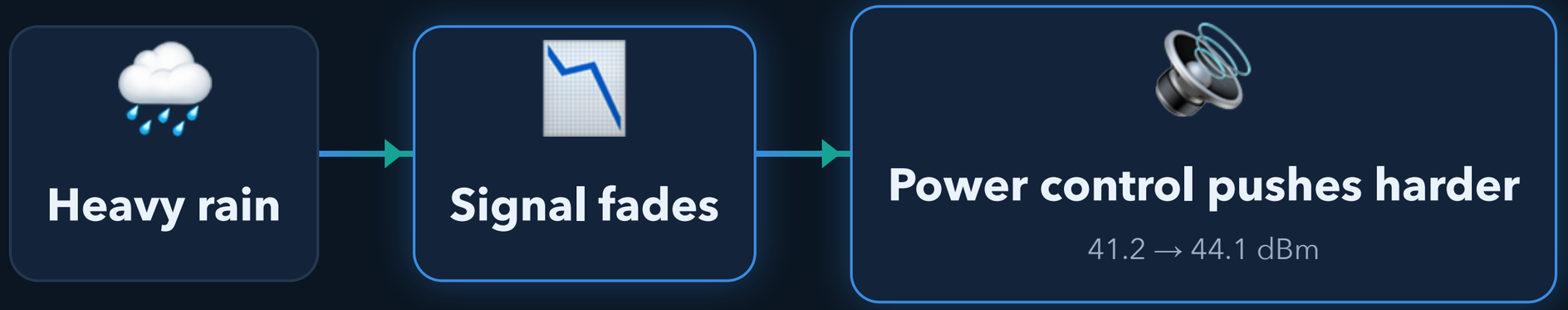
“Wait 2 hours” catches 0 of 12 outages

1 hour 1 Aug 13:00	**1 hour** 8 Aug 18:00	**1 hour** 14 Aug 21:00	**1 hour** 16 Aug 21:00	**1 hour** 17 Aug 20:00	**1 hour** 22 Aug 07:00	**1 hour** 23 Aug 21:00	**1 hour** 24 Aug 03:00
1 hour 26 Aug 08:00	**1 hour** 26 Aug 12:00	**1 hour** 27 Aug 02:00	**1 hour** 30 Aug 21:00				

🙈 Every August outage lasted **exactly one hour**. A delay rule buys quiet by going **blind**.

CAREFUL WHAT YOU FEED A DETECTOR

Uplink power is the system's reaction to rain



0.97

correlation of uplink power with rain

Feed it to an outage detector and the detector just learns **"is it raining?"**. It **leaks the answer** instead of finding faults.

THE THREE SHAPES OF TROUBLE

Spike, level shift, drift

📍

Spike

One odd reading, then back to normal.



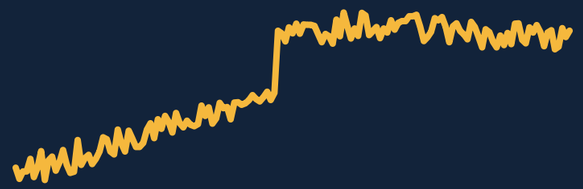
solar-array dropout, 18 Aug

✅ **a limit can catch it (with context)**

🪜

Level shift

Jumps to a new normal and stays there.



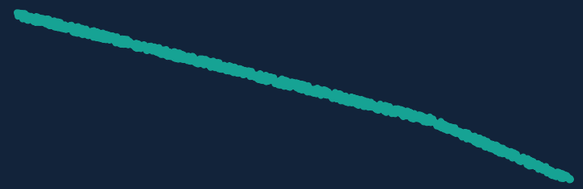
amplifier temp, +2.2 °C in May

❌ **fixed limit fires 0 times**

🐢

Drift

Slides slowly. No single day looks strange.



battery capacity, 3 years

❌ **a 95 Ah limit fires 0 times**

YOUR TURN · 60 MINUTES

Tame the wolves 🐺

1. Open **W3D1-exercise** in Colab, save a copy
2. **Part A:** count the wolves, then add context
3. **Part B:** let the data choose the limit
4. **Part C:** rain, delays, leaks and the 3 shapes



14 checks to pass

13 tasks + 🌶️ the pepper task

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

🚨 **A limit is a model.** Ask where every number came from.

🧠 **Add context, don't widen.** Eclipse, recharge and rain are normal states, not faults.

🪜 **Fixed limits catch spikes**, not level shifts or slow drift.

Next 👉 **Control Charts and Residuals**: a normal that moves with the data.